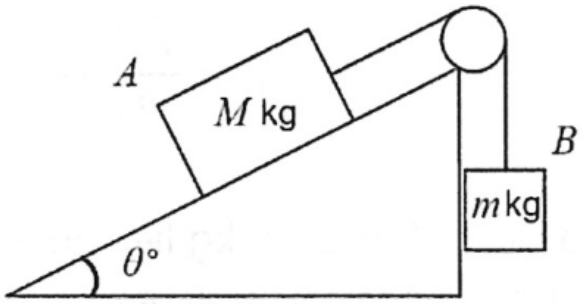
# Forces: Connected Particles on Inclined Planes

Questions 1–16

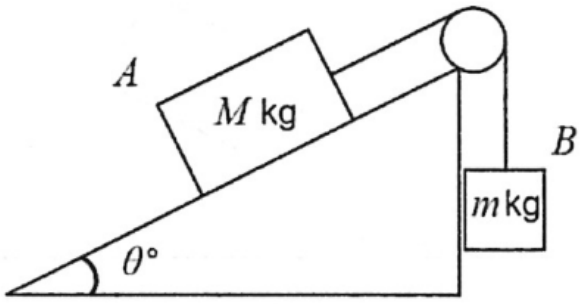
**Question 1. The following question use** $g = 9.8$ ms$^{-2}$ A particle $A$ has mass $M$ kg and is on a smooth slope which is inclined with an angle of $\theta°$. It is being pulled up the slope by a light, inextensible string which is parallel to the slope and passes over a small, smooth, fixed pulley. A second particle $B$ of mass $m$ kg hangs from the pulley freely with the string taut. The system is released from rest.




Find the tension in the string and the acceleration in the system, and state the direction that particle $A$ is moving, given:

a) $M = 1.5, m = 3.9, \theta = 30$
b) $M = 2.1, m = 0.5, \theta = 60$
c) $M = 2, m = 2, \theta = 40$
d) $M = 40, m = 20, \theta = 30$

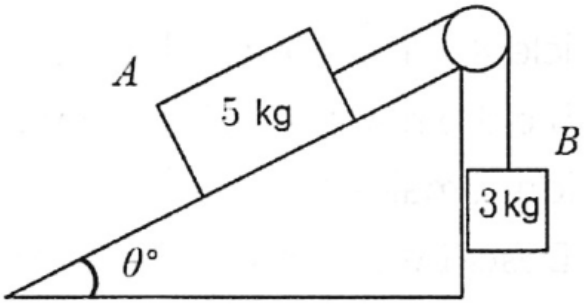
**Question 2.** A particle $A$ has mass $M$ kg and is on a rough slope which is inclined with an angle of $\theta°$. It is being pulled up the slope by a light, inextensible string which is parallel to the slope and passes over a small, smooth, fixed pulley. A second particle $B$ of mass $m$ kg hangs from the pulley freely with the string taut. The coefficient of friction between the slope and the particle is $\mu$. The tension in the string is $T$ N.




Given that the system is released from rest:

a) $M = 2, m = 5, \theta = 30$ and $\mu = 0.85$, find $T$ and the acceleration (and direction) of $A$
b) $M = 7, \theta = 40, \mu = 0.7, a = 0.4$ (up the slope), find $m$ and $T$
c) $M = 7, m = 3, \theta = 60, a = 0.1$ (down the slope), find $T$ and $\mu$
d) $M = 2k, m = k$, where $k$ is a positive constant, $\theta = 30, \mu = 0.5$, find $a$ and give the direction of motion.

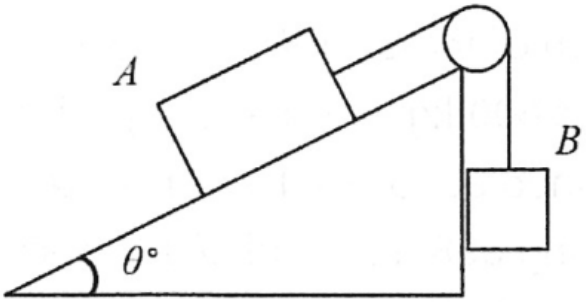
**Question 3.** A box $A$ has mass 5 kg and is on a rough slope which is inclined with an angle of $\theta°$ where $\tan\theta = \frac{4}{3}$. A light, inextensible string connects box $A$ to box $B$ which has mass 3 kg. The string is parallel to the slope and passes over a small, smooth, fixed pulley. Box $B$ hangs from the pulley freely with the string taut.




The system is released from rest and the box moves down the slope at 0.8 $\text{ms}^{-2}$.

a) Find the coefficient of friction between the slope and the box.
b) Find the distance travelled by the system in the first 5 seconds, stating any assumptions you have made.
c) Explain what it means for the string to be modelled as light and inextensible.

**Question 4.** A box $A$ has mass $m$ kg and is on a rough slope which is inclined with an angle of $\theta°$ where $\tan\theta = \frac{3}{4}$. It is being pulled up the slope by a light, inextensible string which is parallel to the slope and passes over a small, smooth, fixed pulley. A second box $B$ of mass $2m$ kg hangs from the pulley freely with the string taut.



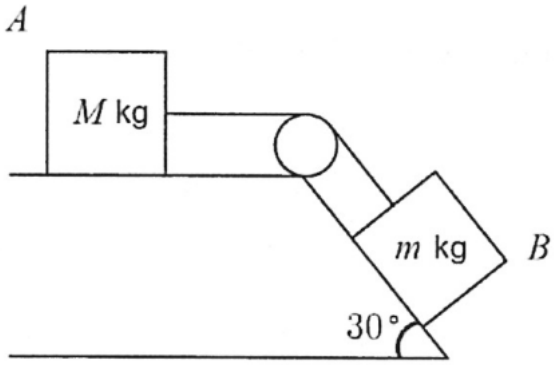

The coefficient of friction between the slope and the box is 0.4. The system is released from rest.

a) Find the tension in the string in terms of $m$

Given that box $B$ starts 1.5 m above the ground and assuming box $A$ does not reach the pulley. Find

b) the time it takes for box $B$ to reach the ground.
c) the total time box $A$ spends moving from its initial position to when it stops.

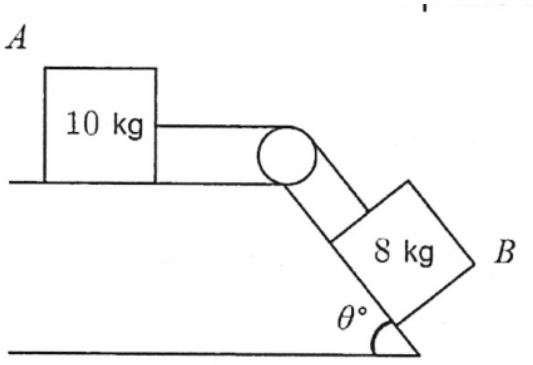
**Question 5. The following questions use** $g = 10$ ms$^{-2}$ A block $A$ has mass $M$ kg and is held at rest on a horizontal rough plane. A second block $B$ of mass $m$ kg lies on a smooth inclined plane with an angle of 30°. Blocks $A$ and $B$ are connected by a light, inextensible string which passes over a small, smooth, fixed pulley and is parallel to the surface. The coefficient of friction between block $A$ and the plane is $\mu$. The tension in the string is given by $T$ N.




The system is released from rest. The acceleration of the system is $a$ ms$^{-2}$. Given

a) $M = 10, m = 3, \mu = 0.2$, find the value of $a$ and $T$
b) $M = 5, m = 4, a = 1.2$, find the value of $T$ and $\mu$
c) $M = 0.5, m = 2, T = 3$, find the value of $a$ and $\mu$
d) $a = 2, \mu = 0.4, T = 12$ find the value of $m$ and $M$

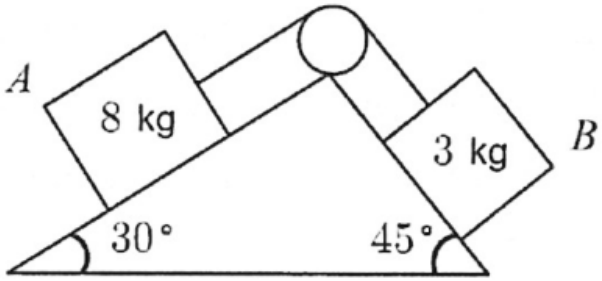
**Question 6.** A block $A$ has mass 10 kg and lies on a horizontal rough plane. A second block $B$ of mass 8 kg lies on a rough inclined plane with an angle of $\theta°$ where $\tan\theta = \frac{4}{3}$. Blocks $A$ and $B$ are connected by a light, inextensible string which passes over a small, smooth, fixed pulley. The coefficient of friction between block $A$ and the plane is 0.2. The coefficient of friction between block $B$ and the plane is 0.4




The system is released from rest and $A$ travels towards the pulley. Find

a) the acceleration of the system.
b) the tension in each of the strings.

**Question 7.** A block $A$ has mass 8 kg and is on a rough slope which is inclined with an angle of 30°. A second block $B$ has mass 3 kg and is on a smooth slope which is inclined with an angle of 45°. Blocks $A$ and $B$ are connected by a light, inextensible string which passes over a small, smooth, fixed pulley.



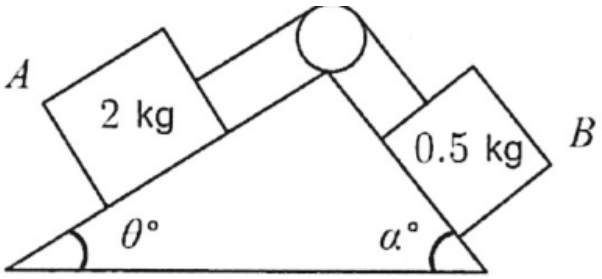

The coefficient of friction between block $A$ and the plane is 0.05. The system is released from rest.

a) Find the acceleration of the system.
b) State the direction in which the blocks are moving.
c) Find the tension in the string.

Block $B$ is replaced by block $C$ with mass $m$ kg and when released it remains in equilibrium.

d) Find the value of $m$

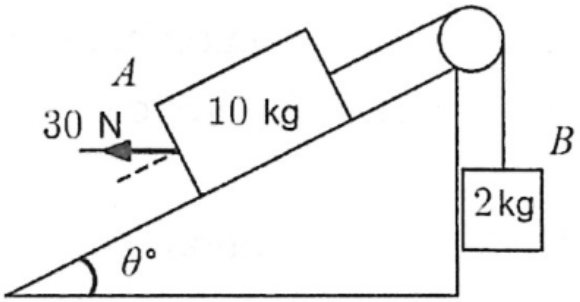
**Question 8.** A block $A$ has mass 2 kg and is on a rough slope which is inclined with an angle of $\theta°$ where $\tan\theta = \frac{3}{4}$. A second block $B$ has mass 0.5 kg and is on a rough slope which is inclined with an angle of $\alpha°$ where $\tan\alpha = \frac{4}{3}$. Blocks $A$ and $B$ are connected by a light, inextensible string which passes over a small, smooth, fixed pulley. The rope is parallel to the slopes as shown in the diagram below.




The coefficient of friction between block $A$ and the plane on which it lies is twice the size of the coefficient of friction between block $B$ and the plane on which it lies.
The system is released from rest and $A$ travels with acceleration 0.1 $\text{ms}^{-2}$ away from the pulley. Find

a) the coefficient of friction between block $A$ and the plane.
b) the tension in the string.

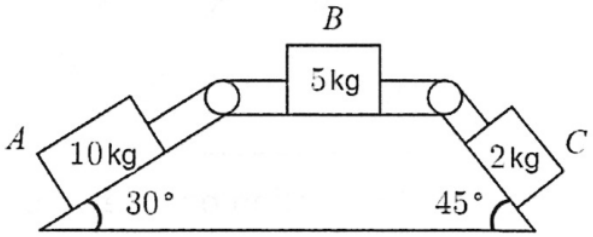
**Question 9. The following questions use** $g = 9.8$ ms$^{-2}$ A box has mass 10 kg is being pulled down a rough slope which is inclined with an angle of $\theta°$ where $\tan\theta = \frac{5}{12}$ by a light, inextensible, horizontal string which has tension 30 N. The particle is also attached to another light, inextensible string which passes over a small, smooth, fixed pulley to a second box of mass 2 kg which hangs from the pulley freely with the string taut.




The acceleration of the system is 0.4 ms$^{-2}$.

a) Find the tension in the string connecting the two boxes.
b) Show that the value of the coefficient of friction is approximately 0.519
c) State any modelling assumptions you have used.

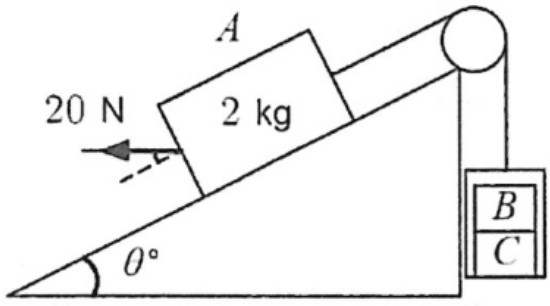
**Question 10.** Box $A$ with mass 10 kg is on a rough plane inclined with an angle of 30° and is connected by a light inextensible string which passes over a small, smooth, fixed pulley to box $B$. Box $B$ with mass 5 kg is on a rough horizontal plane slope and is connected by a light inextensible string which passes over a small, smooth, fixed pulley to box $C$. Box $C$ has mass 2 kg lies on a rough plane which is inclined with an angle of 45° to the horizontal.




The coefficient of friction between each of the boxes and the planes are $\mu_A = 0.1, \mu_B = 0.4$ and $\mu_C = 0.2$. The system is released from rest and box $A$ slides down the plane. Find

a) the acceleration of the system
b) the tension in the string between box $A$ and $B$
c) the tension in the string between box $B$ and $C$

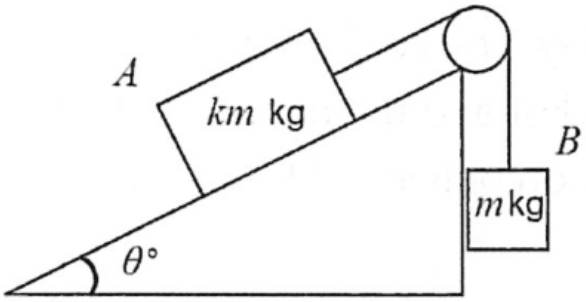
**Question 11. The following questions use** $g = 10$ ms$^{-2}$ A box $A$ has mass 2 kg and is on a rough slope which is inclined with an angle of $\theta°$ where $\tan\theta = \frac{3}{4}$. The box is being pulled by a horizontal force of 20 N. Box $A$ is connected to a light, inextensible string which is parallel to the slope and passes over a small, smooth, fixed pulley to a light container. The container has two boxes stacked inside, with box $B$ sitting directly on top of box $C$. Box $B$ has mass 1 kg and box $C$ has mass 3 kg.




The coefficient of friction between the slope and the box is 0.5

a) Find the acceleration of the system.
b) Find the tension in the string.
c) Find the force exerted on $C$ by $B$
d) Explain how the system changes if the container is no longer modelled as light.

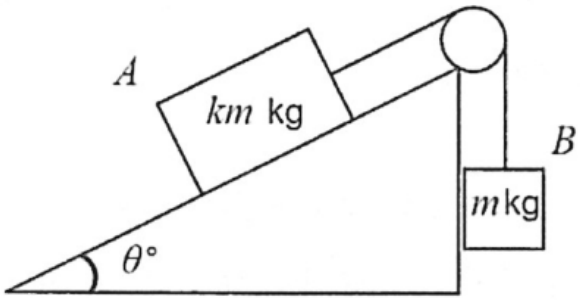
**Question 12. The following questions use exact values of** $g$ A crate $A$ has mass $km$ kg where $k$ is a positive constant. It lies on a rough slope which is inclined with an angle of $\theta°$ where $\tan\theta = \frac{5}{12}$. It is connected by a rope which is parallel to the slope and passes over a small, smooth, fixed pulley. A second box $B$ of mass $m$ kg hangs with the string taut from the same rope.




The coefficient of friction between the slope and the box is $\frac{1}{12}$. The system is released from rest and crate $A$ moves down the slope.

a) Show that $a = \left(\frac{4k-13}{13k+13}\right) g$
b) State any modelling assumptions you have made.

**Question 13.** A particle $A$ has mass $km$ kg and is on a rough slope which is inclined with an angle of $\theta°$ where $\tan\theta = \frac{7}{24}$. It is being pulled up the slope by a light, inextensible string which passes over a small, smooth, fixed pulley. A second box $B$ of mass $m$ kg hangs from the pulley freely with the string taut. The coefficient of friction between the slope and the box is $\mu$




The system is released from rest and particle $A$ moves 0.5 metres up the slope in 2 seconds.

a) Find the acceleration of particle $A$.

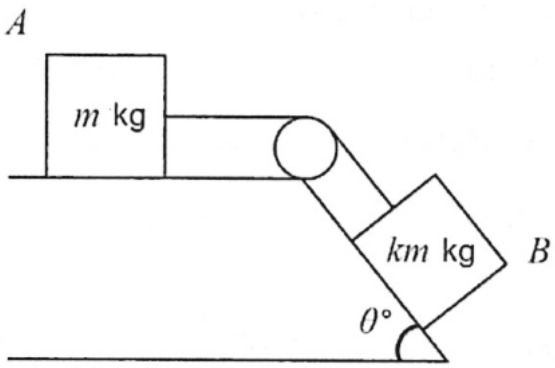
b) Show that the tension in the string is $\left(g - \frac{1}{4}\right) m$

Given that $k = 0.05$

c) Find the value of $\mu$ in terms of $g$

d) State any modelling assumptions you have made.

**Question 14.** A block $A$ has mass $M$ kg and is held at rest on a horizontal rough plane. A second block $B$ of mass $m$ kg lies on a smooth inclined plane with an angle of $\theta°$ where $\tan\theta = \frac{3}{4}$. Blocks $A$ and $B$ are connected by a light, inextensible string which passes over a small, smooth, fixed pulley and is parallel to the surface. The coefficient of friction between block $A$ and the plane is $\mu$

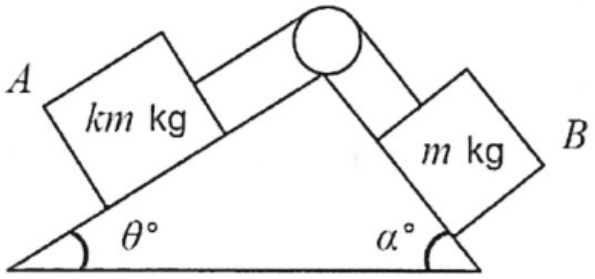



The system is released from rest. The acceleration of the system is $a$ ms$^{-2}$.

a) Show that $a = \frac{(3k-5\mu)g}{5k+5}$

b) Find the range of values of $k$ for which the system will be moving.

**Question 15.** A block $A$ has mass $km$ kg and is on a rough slope which is inclined with an angle of $\theta°$. A second block $B$ has mass $m$ kg and is on a rough slope which is inclined with an angle of $\alpha°$. Blocks $A$ and $B$ are connected by a light, inextensible string which passes over a small, smooth, fixed pulley.



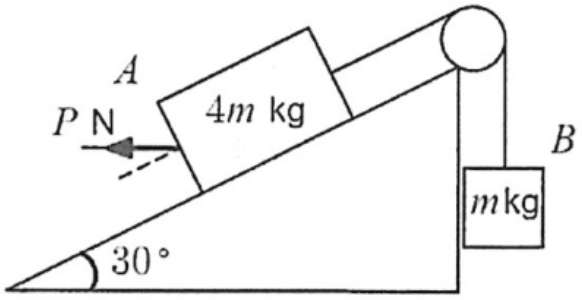

The coefficient of friction between block $B$ and the plane on which it lies is $\mu$. The coefficient of friction between block $A$ and the plane on which it lies is 3 times larger.
The system is released and is at the point of block $B$ sliding down the slope.

a) Show that $k = \frac{\sin\alpha - \mu\cos\alpha}{\sin\theta + 3\mu\cos\theta}$

Block $A$ is 10% heavier than block $B$.

b) In the case that $\theta = 30$ and $\alpha = 40$, find the value of $\mu$ giving your answer to 3 significant figures.

**Question 16.** A box $A$ has mass $4m$ kg and is being pulled along a rough slope which is inclined with an angle of $30°$ by a force of $P$ N which is horizontal, where $P > 0$
The particle is attached to a light, inextensible string which passes over a small, smooth, fixed pulley to a second box of mass $m$ kg which hangs from the pulley freely with the string taut.




The coefficient of friction between box $A$ and the slope is 0.5. The tension in the string which passes over the pulley is $T$ N.
Find the values of $P$ such that the $B$ is moving upwards.